Lesson 10

INTEGRATION

Definition: Integration or anti-derivative of a continuous function $f(x)$, is the reverse process of differentiation.

TYPES OF INTEGRALS :-

1.) INDEFINITE INTEGRAL

If you integrate $f(x)$ without a limit interval on x-values it is called an indefinite integral, and there must be an arbitrary constant.

$$\int f(x)\,dx = \underbrace{F(x) + C}_{\text{Output is function + Constant.}} \quad \text{where } C\text{ - is arbitrary constant.}$$

$$F'(x) = \frac{d}{dx}F(x) = f(x).$$

2.) DEFINITE INTEGRAL

If you integrate $f(x)$ with limit interval on x, say $a \leq x \leq b$, it is called definite integral, and the output is just a real value.

$$\int_a^b f(x)\,dx = \underbrace{F(b) - F(a)}_{\text{output is a real value.}} \quad , \text{ where } F'(x) = \frac{d}{dx}F(x) = f(x)$$

3.) INFINITE INTEGRAL / IMPROPER INTEGRAL

It is a definite integral for which limits may involve $\pm\infty$.

Examples:

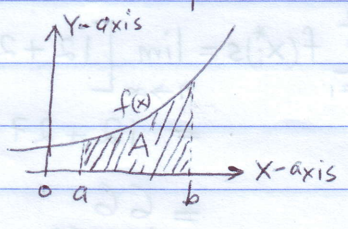
(a) $\int_{-\infty}^{b} f(x)\,dx$ OR (b) $\int_{a}^{\infty} f(x)\,dx$ OR (c) $\int_{-\infty}^{\infty} f(x)\,dx$

*We will learn how to solve such integrals later in the course.

DEFINITE INTEGRAL

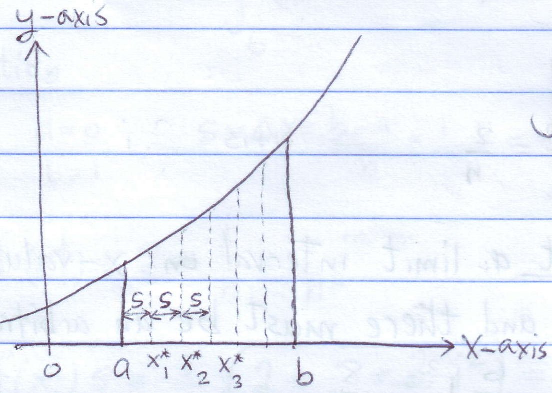
A definite integral can be interpreted as area under the curve.

$$\int_a^b f(x)\,dx = A$$



DEFINITE INTEGRAL USING RIEMANN SUMS

The limit definition of definite integral of $f(x)$ on interval $a \le x \le b$ Can be Computed using Riemann sums: (This Compute exact area under curve)



$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\,s \qquad s = \Delta x$$

Where $s = \Delta x = \frac{b-a}{n}$ (interval step)

$$x_i^* = a + is, \quad i = 1, 2, 3, \ldots, n$$

FORMULAS FOR SUMS

(1) $\sum_{i=1}^{n} c = nc$; (2) $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$; (3) $\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$; (4) $\sum_{i=1}^{n} i^3 = \frac{n^2(n+1)^2}{4}$

Example 1

Use limit definition of integral and Riemann Sum to evaluate the integral

$$\int_1^4 (3x^2+1)\,dx$$

Solution

Here $a=1$, $b=4$, $s = \Delta x = \frac{b-a}{n} = \frac{4-1}{n} = \frac{3}{n}$; $x_i^* = a + is = 1 + \frac{3}{n}i$; $f(x) = 3x^2+1$

$$f(x_i^*) = 3\left(1+\frac{3}{n}i\right)^2 + 1 = 3\left(1 + \frac{6}{n}i + \frac{9}{n^2}i^2\right) + 1 = 4 + \frac{18}{n}i + \frac{27}{n^2}i^2$$

$$f(x_i^*)\,s = \frac{3}{n}\left[4 + \frac{18}{n}i + \frac{27}{n^2}i^2\right] = \frac{12}{n} + \frac{54}{n^2}i + \frac{81}{n^3}i^2$$

Riemann Sum:

$$\sum_{i=1}^{n} f(x_i^*)\,s = \sum_{i=1}^{n}\frac{12}{n} + \frac{54}{n^2}\sum_{i=1}^{n} i + \frac{81}{n^3}\sum_{i=1}^{n} i^2$$

$$= \frac{12}{n}\cdot n + \frac{54}{n^2}\cdot\frac{n(n+1)}{2} + \frac{81}{n^3}\cdot\frac{n(n+1)(2n+1)}{6}$$

$$= 12 + 27\left(1+\frac{1}{n}\right) + \frac{81}{6}\cdot\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)$$

$$\int_1^4 (3x^2+1)\,dx = \lim_{n\to\infty}\sum_{i=1}^{n} f(x_i^*)\,s = \lim_{n\to\infty}\left[12 + 27\left(1+\frac{1}{n}\right) + \frac{81}{6}\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)\right]$$

Note: $n\to\infty$ $\left(\frac{1}{n}\to 0\right)$:

$$= 12 + 27(1+0) + \frac{81}{6}(1)(2)$$

$$= 66$$

: Check Ans using CALCULUS:

Example 2

Use limit definition of integral to evaluate the integral

$$\int_0^2 (x^3-4)\,dx$$

Solution

Here $a=0$, $b=2$; $s=\Delta x=\frac{b-a}{n}=\frac{2-0}{n}=\frac{2}{n}$, $x_i^*=a+is=\frac{2}{n}i$

$$f(x_i^*)=\left(\frac{2}{n}i\right)^3-4=\frac{8}{n^3}i^3-4$$

$$f(x_i^*)s=\frac{2}{n}\left(\frac{8}{n^3}i^3-4\right)=\frac{16}{n^4}i^3-\frac{8}{n}$$

Riemann Sum:

$$\sum_{i=1}^{n} f(x_i^*)s=\sum_{i=1}^{n}\left[\frac{16}{n^4}i^3-\frac{8}{n}\right]=\frac{16}{n^4}\sum_{i=1}^{n}i^3-\sum_{i=1}^{n}\frac{8}{n}$$

$$=\frac{16}{n^4}\cdot\frac{n^2(n+1)^2}{4}-\frac{8}{n}\cdot n$$

$$=4\left(1+\frac{1}{n}\right)^2-8$$

$$\int_0^2 (x^3-4)\,dx=\lim_{n\to\infty}\sum_{i=1}^{n} f(x_i^*)s$$

$$=\lim_{n\to\infty}\left[4\left(1+\frac{1}{n}\right)^2-8\right]$$

$$=4(1+0)^2-8$$

$$=4-8$$

$$=-4$$

Check ANS, using Calculus: $\int_0^2 (x^3-4)\,dx=\frac{x^4}{4}-4x\Big|_0^2=\frac{16}{4}-8=-4$ CORRECT!!

Example 3

Use limit definition of integral to evaluate

$$\int_0^1 (2-x+2x^2)\,dx$$

Solution

Here $a=0$, $b=1$; $s=\Delta x=\frac{b-a}{n}=\frac{1-0}{n}=\frac{1}{n}$; $x_i^*=a+is=0+\frac{1}{n}i=\frac{i}{n}$

$$f(x_i^*)=2-\frac{i}{n}+\frac{2}{n^2}i^2$$

$$f(x_i^*)s=\frac{1}{n}\left(2-\frac{i}{n}+\frac{2}{n^2}i^2\right)=\frac{2}{n}-\frac{i}{n^2}+\frac{2}{n^3}i^2$$

Riemann Sums:

$$\sum_{i=1}^{n} f(x_i^*)s=\sum_{i=1}^{n}\left(\frac{2}{n}-\frac{i}{n^2}+\frac{2}{n^3}i^2\right)$$

$$=\frac{2}{n}\cdot n-\frac{1}{n^2}\sum_{i=1}^{n} i+\frac{2}{n^3}\sum_{i=1}^{n} i^2$$

$$=2-\frac{1}{n^2}\cdot\frac{n(n+1)}{2}+\frac{2}{n^3}\cdot\frac{n(n+1)(2n+1)}{6}$$

$$=2-\frac{1}{2}\left(1+\frac{1}{n}\right)+\frac{1}{3}\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)$$

$$\int_0^1 (2-x+2x^2)\,dx=\lim_{n\to\infty}\sum_{i=1}^{n} f(x_i^*)s=\lim_{n\to\infty}\left[2-\frac{1}{2}\left(1+\frac{1}{n}\right)+\frac{1}{3}\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)\right]$$

$$=2-\frac{1}{2}(1+0)+\frac{1}{3}(1+0)(2+0)$$

$$=2-\frac{1}{2}+\frac{2}{3}$$

$$=\frac{13}{6}$$

Check:

$$\int_0^1 (2-x+2x^2)\,dx=\left[2x-\frac{x^2}{2}+\frac{2}{3}x^3\right]_0^1=2-\frac{1}{2}+\frac{2}{3}=\frac{13}{6}$$ ✓ YES!!